Sep 9 Recap: Joint $f_{XY}(x,y)$ Marginals $f_X(x)$, $f_Y(y)$

Computing probs

Independence of RVs

A, B: events are mutually indep. if $P(A \cap B) = P(A)P(B)$

X, Y: RVs jointly distributed

Events defined using $X \to X \le x \quad x \in \mathbb{R}$

$$\bigcup_{n=1}^{\infty} \left(X \le x - \tfrac{1}{n}\right) = X < x \; ?$$

$x - \varepsilon$ as $\varepsilon \to 0$

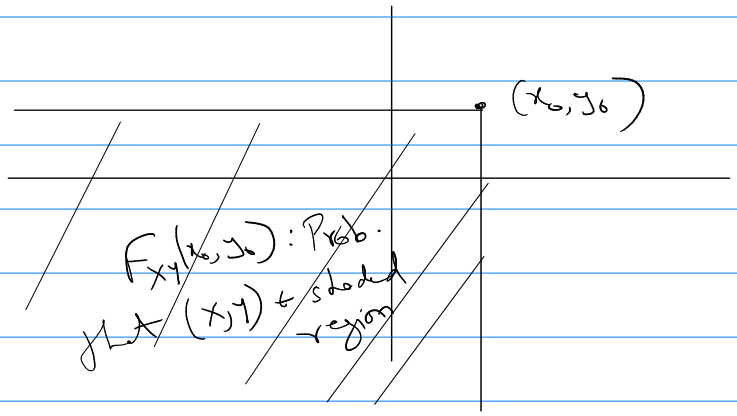
Events defined using $Y \to Y \le y \quad y \in \mathbb{R}$

Def: X & Y are indep. if $P(X \le x, Y \le y) = P(X \le x)P(Y \le y)$ for all x, y.

$P(X \le x)$: CDF of X, $F_X(x)$ $\quad$ $P(Y \le y)$: CDF of Y, $F_Y(y)$

Joint CDF of X & Y:

$$F_{XY}(x,y) = P(X \le x, Y \le y)$$



Jointly continuous

$$f_{XY}(x,y) = \frac{\partial}{\partial x \partial y} F_{XY}(x,y)$$

Discrete: X & Y are indep if $\leftarrow P(X=x, Y=y)$

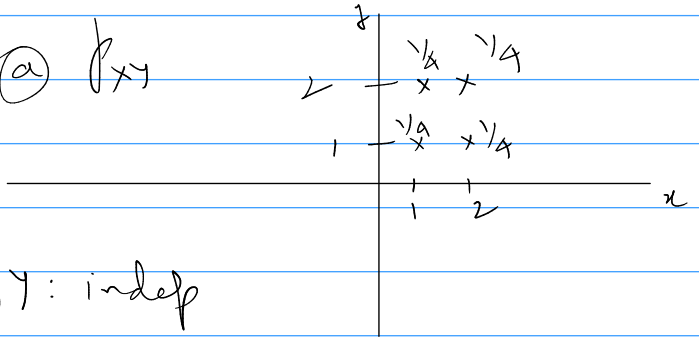
joint PMF factors $\leftarrow p_{XY}(x,y) = p_X(x)\, p_Y(y)$ for all x, y.

($p_X(x) = P(X=x)$, $p_Y(y) = P(Y=y)$)

Jointly continuous X & Y are indep. if

joint PDF factors $\leftarrow f_{XY}(x,y) = f_X(x)\, f_Y(y)$ for all x, y.

Ex: (1) (a) p_{XY}



	1	2	
1	1/4	1/4	1/2
2	1/4	1/4	1/2
	1/2	1/2	

X, Y: indep

(b)

indep

y \ x	0	1	
0	$(1-p)^2$	$p(1-p)$	$1-p$
1	$p(1-p)$	p^2	p
	$1-p$	p	

$p \in (0,1)$

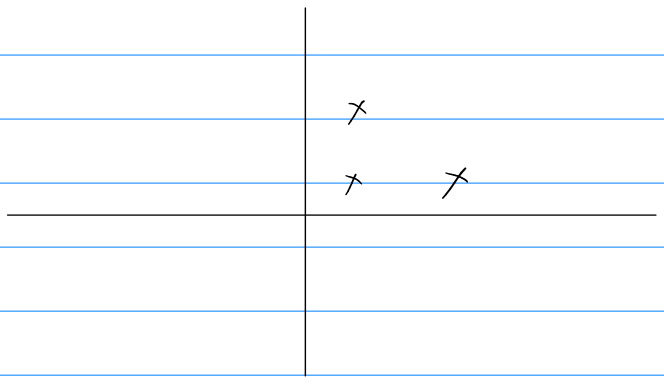
(c)

y \ x	0	1
0	p_{00}	p_{01}
1	p_{10}	p_{11}

may not be indep

Ex: $p_{01} = 0$, $p_{00} \neq 0$, $p_{10} \neq 0$, $p_{11} \neq 0$

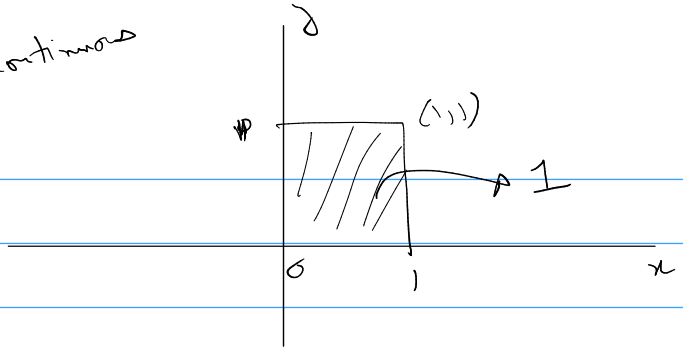
(d)



not indep.

Ex: Jointly continuous

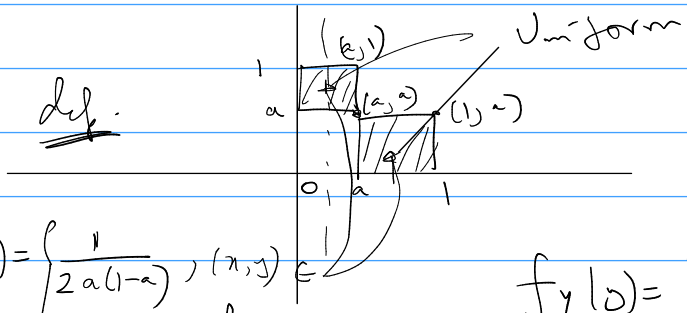
(a)



indep.

$X, Y \sim \text{Uniform}[0,1]$

(b)



def.

$$f_{XY}(x,y) = \begin{cases} \frac{1}{2a(1-a)}, & (x,y) \in \\ 0, & \text{else} \end{cases}$$

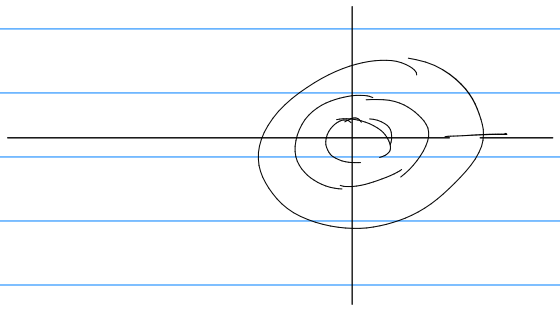
$$f_X(x) = \begin{cases} \frac{1}{2a}, & 0 < x < a \\ \frac{1}{2(1-a)}, & a < x < 1 \\ 0, & \text{else} \end{cases}$$

$$f_Y(y) = \begin{cases} 1/2a, & 0 < y < a \\ \frac{1}{2(1-a)}, & a < y < 1 \end{cases}$$

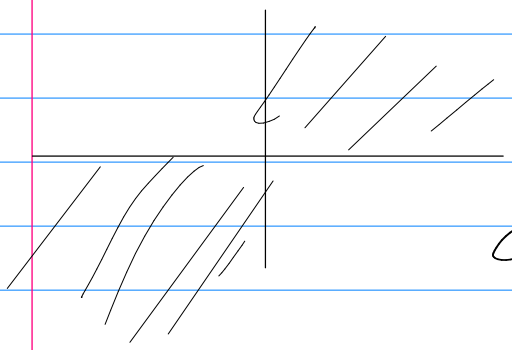
$a = 1/2$ $X, Y \sim \text{Uniform}[0,1]$

(c) $X, Y \sim$ PDF $\frac{1}{\sqrt{2\pi}} e^{-u^2/2}$, $u \in \mathbb{R}$

X, Y: indep. Joint PDF: $f_{XY}(x,y) = \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}}$, all x, y



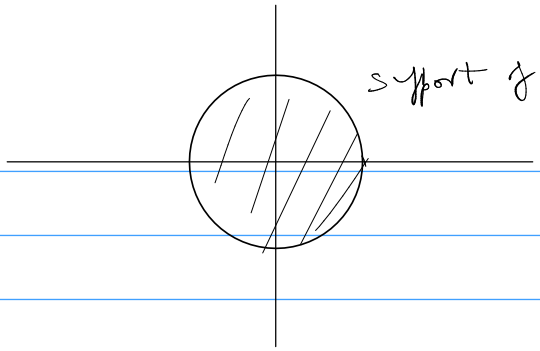
(b) Joint PDF



$$f_{XY}(x,y) = \begin{cases} \frac{1}{2\pi} e^{-\frac{(x^2+y^2)}{2}}, & x,y > 0, \\ & x,y < 0 \\ 0, & \text{else} \end{cases}$$

Check: $X, Y \sim$ marginal PDF $\frac{1}{\sqrt{2\pi}} e^{-u^2/2}$, $u \in \mathbb{R}$

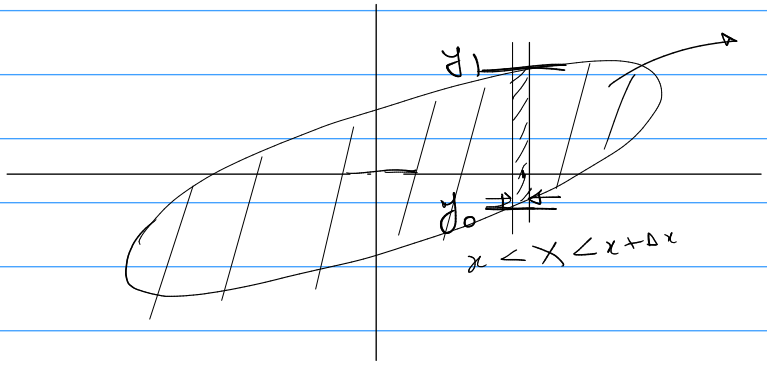
Ex:



not indep.

→ indep. simplifies models

Continuous: Conditional PDFs



$P(A|B)$

$P(B) > 0$

Define Event E: using (X,Y) s.t. $P(E) > 0$.

$X, Y \mid E$

$$f_{XY|E}(x,y) = \begin{cases} \frac{f_{XY}(x,y)}{P(E)}, & (x,y) \in E \\ 0, & \text{else.} \end{cases}$$

$$P(x < X < x+\Delta x,\ y < Y < y+\Delta y) \approx f_{XY}(x,y)\,\Delta x\,\Delta y$$

$$P(x < X < x+\Delta x,\ y < Y < y+\Delta y \mid E) \approx \begin{cases} \frac{f_{XY}(x,y)\Delta x \Delta y}{P(E)}, & (x,y) \in E \\ 0, & \text{else} \end{cases}$$